

Case Study Rubric

DS 4002 – Spring 2026 - Instructors: Karsten Siller; TA: Cole Whittington

Due: TBD

Submission format:

- **Upload link to github repo and PDF file to Canvas**

Individual Assignment

General Description: Submit to canvas a link to your github repo and PDF file.

Why am I doing this? This case study is an opportunity to showcase your technical skills as well as conceptual skills in a unified project applicable to a real world issue. This case study will represent a hands-on scenario similar to one you may be presented in a professional environment. This assignment is designed to help you practice communicating your work, structuring a project, and thinking critically about how data science solutions operate beyond the classroom.

- Course Learning Objective: communicating your work, structuring a project, and thinking critically about how data science solutions operate beyond the classroom

What am I going to do?

By this point in the course, you've built a foundation of technical and conceptual skills across data cleaning, modeling, evaluation, and communication. This case study gives you the opportunity to bring those pieces together into a cohesive, independently-driven project modeled after a real-world data science scenario. You will take the structure of a professional workflow—framing a problem, preparing data, designing a modeling approach, and communicating results—and adapt it to the context of medical image classification.

Your goal is to produce a complete deliverable that demonstrates your ability to translate a complex problem into a clear, well-reasoned data science solution. This includes:

- A written component that explains your process and findings (PDF format)
- a GitHub repository containing the materials needed for someone else to understand and reproduce your work.

Tips for success:

- Stay focused on clarity over complexity. A well-explained, well-structured workflow is more valuable than an overly advanced model.
- Document your decisions as you go so your final deliverable clearly shows how and why you built your pipeline the way you did.
- Test your work early and often; small checks on data, code, and outputs will save you time and help ensure your final results are reliable.

How will I know I have Succeeded? You will meet expectations on this case study when you follow the criteria in the rubric below:

Formatting	Submit each component listed in the rest of this rubric as advised below. ● Repository – A GitHub repo (and cloud storage folder if necessary) containing all materials ○ Submit a link to the repo ○ Everything is contained in the repo or linked to it if appropriate. ○ Use pdf format when possible ○ For code and data products use the appropriate format for whatever it is ● Written Portion ○ Submit the written portion as a PDF file AND ○ Attach it to the repository ● Data and Code ○ Submit code created for all portions in a GitHub repository ○ Include any additional data that was used in GitHub if necessary ○ The GitHub should be titled “CS-[insert first & last name]” ● References ○ References should be included on a separate page at the end of the Written Portion PDF file ○ IEEE citation style
Repository	The GitHub repository for this case study should contain ALL materials used for this project, including: ● A folder labeled DATA with all data sources used in the project ● Folder labeled OUTPUT with all outputs of the project, including ○ Graphs, tables, etc. ○ Written Portion in PDF format with References ○ Other deliverables ● Folder labeled SCRIPTS with all code scripts used for the project ● LICENSE.md file

	• README.md file outlining steps to reproduce results and all files in the repository
Written Portion	Discuss your interpretation of the study and thought process by answering the questions below: • Summarize the problem and its importance (One paragraph) ○ In a short paragraph, clearly describe the real-world problem your case study addresses and explain why it matters. This should establish the context for your work and show that you understand the broader significance of the task. • Explain your plan for completing the deliverable (Two paragraphs) ○ In a brief paragraph, outline the approach you took to meet the requirements of the case study. ○ Include a simple graphic (e.g., workflow diagram, pipeline sketch, or step-by-step outline) that visually represents the analysis plan you followed. This section should show how you structured your work and why you chose that structure. ○ Outline challenges you faced and how you overcame them as well as other relevant details about the process in a second paragraph • Present and interpret your results (Two paragraphs) ○ Describe the outcomes of your analysis in complete sentences. Summarize what your model or workflow produced, highlight key findings, and explain what those results mean in the larger context of the problem. Your interpretation should demonstrate that you can connect technical outputs to meaningful conclusions. ○ Discuss next steps, things you would have done differently, and how you plan to improve in the future • Include References for the entire project in this PDF as well
References	At the end of the written portion of this assignment, please include a list of references in IEEE citation style that were NOT in the already given reference materials. For guidance on IEEE Citations, refer to this article: • https://owl.purdue.edu/owl/research_and_citation/ieee_style/ieee_overview.html

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